

Unassessed exercises / problems:

(1) You have developed a new algorithm, called MegaMerge (MM), for joining multiple lists of numbers together then sorting them into ascending order. You believe it does this faster than your existing algorithm called KiloMerge.

(a) State this belief as a testable hypothesis

(b) Explain how it meets the essential criteria of a good hypothesis

(2) Describe three ways in which you can increase the generalisability of an experiment that compares three different online search engines

(3) The research question to be addressed is: “Which hand is it better to use on a computer keyboard, the left or the right?”. This question is asked in the context of the scenario where you have broken one of your wrists – which wrist would you rather break so that you are still able to use a keyboard with one hand?

Design an experiment by defining the factors and describing the experimental process used.

(4) The research question to be addressed is: “Which search engine is best?” Note that this is a very broad topic, so you will need to narrow its scope carefully.

Design an experiment by defining the factors and describing the experimental process used.